

Desiccation Tolerance Studied in the Resurrection Plant *Craterostigma plantagineum*¹

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SYNOPSIS. This review will focus on the acquisition of desiccation tolerance in the resurrection plant *Craterostigma plantagineum*. Molecular aspects of desiccation tolerance in this plant will be compared with the response of non-tolerant plants to dehydration. Unique features of *C. plantagineum* are described like the *CDT-1* (Craterostigma desiccation tolerance gene-1) gene and the carbohydrate metabolism. Abundant proteins which are associated with the desiccation tolerance phenomenon are the late embryogenesis abundant (=LEA) proteins. These proteins are very hydrophilic and occur in several other species which have acquired desiccation tolerance.

INTRODUCTION

Most higher plants are unable to survive desiccation to an air-dried state and only seeds and pollen grains can withstand air dryness for certain periods of time. Pollen grains loose tolerance quickly, whereas seeds can stay for longer in the desiccated state (see also contribution by Walters *et al.*, 2005). However, a small group of vascular angiosperm plants termed “resurrection plants” have also evolved desiccation tolerance, and they can revive from an air-dried state. This is the severest form of water stress, since most protoplasmic water is lost from the cell under these conditions. Resurrection plants are able to withstand severe water loss, and some are even able to equilibrate the leaves with air to 0% (v/v) relative humidity (Gaff, 1971, 1987). The resurrection capacity is normally extended to all tissues of the plant. Resurrection plants are mostly poikilohydrous, which means that their water content adjusts with the relative humidity in the environment. They are able to stay in the dehydrated state until water becomes available and allows them to rehydrate and to resume full physiological activities. During the dehydration process leaves of resurrection plants shrink and curl up. Mature tissue of resurrection plants such as leaves and roots, are able to remain in the air-dried state for months by reaching a quiescent state which is comparable with dormancy in seeds in several aspects. Resurrection plants take immediate advantage of rainfall after dry periods: they resurrect, grow and reproduce before other species can do so (Scott, 2000). Studies have shown that the physiological conditions of the plants are critical for the ability to exhibit desiccation tolerance.

Two types of resurrection plants can be distinguished. One group is classified as poikilochlorophyllous, which means that the plants lose chlorophyll and the thylakoid membranes are at least partially degraded during water loss. Other resurrection plants like, *e.g.*,

Craterostigma plantagineum are homoiochlorophyllous and retain chlorophyll and the intact photosynthetic structures (Tuba *et al.*, 1998), although also changes in the photosynthetic pigments occur (Alamillo and Bartels, 2001). These plants recover rapidly after desiccation and restore photosynthetic activities within 24 hours of rehydration (Bernacchia *et al.*, 1996). The poikilochlorophyllous plants take longer to recover their photosynthetic activity.

Resurrection plants are found in places where substantial rains are seasonal and extremely sporadic. They often grow on rocky outcrops at low to moderate elevations in tropical and subtropical climates (Porembski and Barthlott, 2001). Most resurrection plants have been reported from the southern hemisphere of Africa and from Australia.

Resurrection plants have been identified within the angiosperms both among monocotyledonous and dicotyledonous plants, but no desiccation tolerant gymnosperms or trees have been reported yet. The dicotyledonous plants are represented mainly in the *Scrophulariaceae* and *Myrothamnaceae* families whereas the monocotyledonous are more scattered among different families. A phylogenetic analysis among the *Scrophulariaceae* suggests a clustering of desiccation tolerant plants represented by the genera *Craterostigma* and *Lindernia* (Ramanzadeh *et al.*, 2005). The phylogenetic analysis is a first step towards investigating the evolution of molecular and biochemical characters related to desiccation tolerance.

It has been suggested that desiccation tolerance is connected with a size limitation, since all examples of desiccation tolerant flowering plants do not exceed a certain height (Bewley and Krochko, 1982), perhaps the largest known resurrection plant is the small woody shrub *Myrothamnus flabellifolia* (Sherwin *et al.*, 1998). Most resurrection plants are herbaceous plants.

TEXT

Craterostigma plantagineum a representative of the resurrection plants

This review focuses on the resurrection plant *Craterostigma plantagineum*, a member of the Linderniaceae,

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which is a tribus of the Scrophulariaceae, a large heterogeneous family of polyphyletic origin. Around 170 *Lindernia* species have been reported and most of them are endemic in Africa (Fischer, 2004). The African species mainly occur in specialized habitats like seasonally water-filled rock pools, inselbergs or heavy metal containing soils (Fischer, 2004). The distribution in various ecological niches must have been associated with the invention of adaptive mechanisms. *C. plantagineum* as resurrection plant is an example for this. The aspect which will be considered here is which consequences does the acquisition of desiccation tolerance have on the level of gene expression? The molecular basis of desiccation tolerance is complex and it is not clear yet how and whether mechanisms may vary between different species. So far information on genome sequences for the resurrection plants is very limited, but *C. plantagineum* is the best studied example (Bartels and Salamini, 2001; Bernacchia and Furini, 2004). The available molecular information will be correlated with known genome sequences for the purpose of this review. The study of *C. plantagineum* revealed major changes in specific gene expression patterns and in carbohydrate metabolism characteristic for the dehydration and rehydration stage (Bartels and Salamini, 2001; Bianchi *et al.*, 1991).

Gene expression during dehydration

C. plantagineum is particularly suited for molecular analysis, because desiccation tolerance can be studied in undifferentiated callus cultures and in differentiated plants, which makes it possible to compare gene expression in two genetically identical systems. Desiccation tolerance in callus requires a pretreatment with the plant hormone abscisic acid (ABA) (Bartels *et al.*, 1990). The ABA treatment induces the same genes as dehydration does in the differentiated plant. Changes of gene expression programs have been investigated during different stages of dehydration and rehydration. In *C. plantagineum* the dehydration process is characterized by the activation of many dehydration-induced genes and the accumulation of the respective gene products, which rapidly decline during the rehydration process (Bernacchia *et al.*, 1996). This is different to the moss *T. ruralis* where most changes in gene expression occur during the first hours of rehydration and so-called rehydrins are synthesized (Wood and Oliver, 1999, [see also contribution by M. Oliver and B. Mishler, 2005]). In *T. ruralis* the changes mainly occur on the translational level by selecting different mRNAs for translation from an unchanged pool of mRNAs available for translation independent from the water status of the moss. These observations suggest that different mechanisms must be involved in desiccation tolerance in the moss and higher plants. It will be interesting to know whether this is a general feature distinguishing higher plants and mosses.

Differentially expressed genes were isolated during different stages of dehydration from vegetative tissue and from callus tissue of *C. plantagineum*. It has been

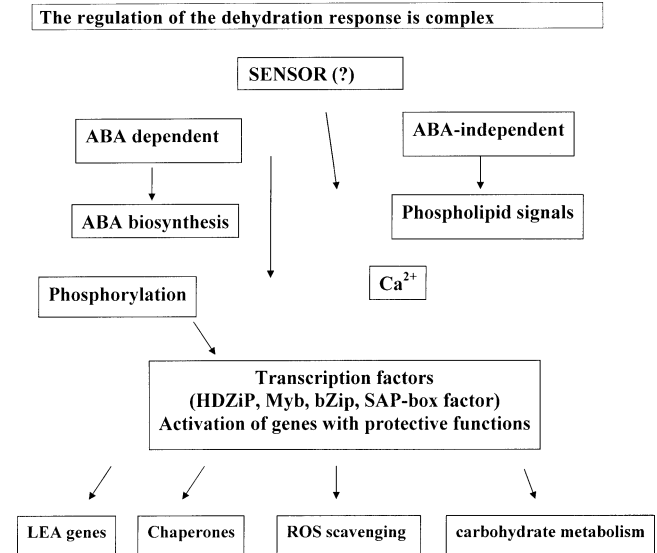


FIG. 1. The diagram indicates the different steps and possible intermediate reactions between signal perception and expression of protective genes.

estimated from these experiments that several hundred genes are differentially expressed during dehydration (Bockel *et al.*, 1998). The DNA sequence analysis of these genes revealed a broad spectrum with diverse functional identities. This is very similar to the non-tolerant plant *Arabidopsis* (Seki *et al.*, 2002; Oono *et al.*, 2003).

Based on conservation of gene contents the following functional categories of dehydration-induced genes in *C. plantagineum* can be distinguished: (I) genes encoding protective proteins including hydrophilins and detoxifying enzymes (see below), (II) genes encoding products with regulatory functions (proteins and RNAs), (III) genes encoding enzymes related to carbohydrate metabolism, (IV) proteins involved in transport of water and other molecules.

Current evidence suggests that multiple mechanisms and several signalling pathways are involved in activating the dehydration response. An indication of the signal cascade is depicted in Figure 1. Unknown is the way in which the stress is perceived. Several pathways in parallel lead to the transcriptional activation of genes encoding protective molecules.

Hydrophilic proteins

The most abundant group of genes which are expressed in response to dehydration are hydrophilic proteins which largely comprise the LEA (=late embryogenesis abundant) proteins. Garay-Arroyo *et al.* (2000) defined hydrophilins which are characterized by a glycine content >6% and a hydrophilicity index >1. Hydrophilins are present in plants, nematodes, fungi and bacteria and accumulate in response to osmotic stress (Garay-Arroyo *et al.*, 2000; Browne *et al.*, 2002). The plant LEA proteins are abundantly expressed in the embryo tissues during seed maturation when the seed

looses water and acquires desiccation tolerance, hence the name LEA. LEA proteins are also expressed in vegetative tissues in response to osmotic stress (Bartels and Sunkar, 2005; Ingram and Bartels, 1996). LEA protein encoding genes are ubiquitously present in plant genomes and have probably a primary role during seed maturation. LEA proteins have been classified into six groups on the basis of conserved sequence motifs (Cuming, 1999). Conserved motifs are likely to define functional domains within the LEA proteins, although the biochemical functions of the protein domains are still unknown. The characteristic expression pattern provides circumstantial evidence for a protective role of the LEA proteins. *In vitro* experiments and analyses of transgenic plants demonstrate that LEA proteins assume a protective role in dehydrating cells. *In vitro* protection assays have shown that selected LEA proteins are able to protect enzyme activities from inactivation caused by water depletion (Rinne *et al.*, 1999; Reyes *et al.*, 2005; Goyal *et al.*, 2005). Ectopic expression of LEA proteins in some transgenic plants or yeast indicated also a correlation between the expression of LEA proteins and stress protection (Imai *et al.*, 1996; Xu *et al.*, 1996; Zhang *et al.*, 2000; Hara *et al.*, 2003; Bartels and Mattar, unpublished). In *C. plantagineum* accumulation of LEA proteins and related hydrophilins have been observed in different cellular compartments like cytoplasm or chloroplasts (Schneider *et al.*, 1993), which again is in agreement with a general protective role, as all cellular structures require protection. Several suggestions have been made how *lea* genes may exert the protective function. LEA proteins could replace water and thus maintain the hydration shell of proteins and other molecules. Another suggestion is that some LEA proteins bind ions and compensate the increasing ionic concentration in dehydrated cells. However, biochemical evidence for these hypotheses is still lacking. LEA proteins may interact with carbohydrates to prevent cellular damage during dehydration.

The dehydrins, one specific group of LEA proteins, undergo posttranslational modification in form of phosphorylation (Alsheikh *et al.*, 2003; Jiang and Wang, 2004; Plana *et al.*, 1991). Phosphorylation may increase the hydrophilicity of the molecules or may allow cellular shuttling as suggested for the rab17 dehydrin of maize by Goday *et al.* (1994).

If the different LEA proteins expressed in vegetative tissues of *C. plantagineum* are surveyed, it is noticeable that they are very abundant and that many different classes are represented (Bartels and Salamini, 2001; Ingram and Bartels, 1996; Ditzer *et al.*, 2001; Rodrigo *et al.*, 2004). Interestingly, one of the LEA-related genes, *CpEdi-9* is more similar to a nematode gene than to known plant LEA-like genes (Rodrigo *et al.*, 2004). The nematode gene has also been correlated with anhydrobiosis in the nematode *Aphelenchus avenae* (Browne *et al.*, 2002). This supports a general role for LEA proteins as dehydration protectant across species and indicates that LEA proteins must have

been functionally successful during evolution. LEA proteins may have evolved in the genome of *C. plantagineum* by re-shuffling sequence fragments encoding hydrophilic protein segments. Desiccation tolerance is an adaptation to extreme environmental conditions, which may exert genotoxic stress leading to abundant expression of hydrophilic proteins as a survival mechanism. It will be interesting to determine whether common regulatory motifs and modules can be identified in the regulatory regions of the stress genes. A comparison of promoter elements and sequences of the known *C. plantagineum* LEA genes did so far not reveal common promoter structures, but this may require better comparative bioinformatics tools and the availability of many more genome sequences than are available at present. It should, however, be mentioned that some *cis* regulatory motifs like dehydration-response or ABA-response elements are often present in the promoter regions but it is difficult to predict their functions.

CDT (=Craterostigma desiccation tolerance) genes are unique to *C. plantagineum*

There are several lines of evidence that ABA has a central role in the acquisition of desiccation tolerance in *C. plantagineum*. The importance of ABA as signaling molecule in osmotic stress responses has been convincingly demonstrated by multiple mutants in *Arabidopsis thaliana* (Leung and Giraudat, 1998; Himmelbach *et al.*, 2003). A straightforward mutant analysis is not possible in *C. plantagineum*, because the genome is not diploid. The role of ABA in the acquisition of desiccation tolerance in *C. plantagineum* is supported by two observations. Firstly, ABA is required to render callus tissue desiccation tolerant; callus which has not been treated with ABA does not become desiccation tolerant (Bartels *et al.*, 1990). Secondly, ABA levels increase during dehydration, many of the dehydration-induced genes can also be induced by ABA and contain ABA response elements in their promoters (Bartels and Salamini, 2001). The strict requirement of ABA to induce desiccation tolerance in callus was utilized to design a screen for a dominant mutation by using an activation tagging approach (Furini *et al.*, 1997; Smith *et al.*, unpublished). The result of the first screen was the isolation of the *CDT-1* gene. The activation and constitutive expression of *CDT-1* leads to callus which does not require ABA treatment to express desiccation tolerance and in which desiccation-responsive genes like *lea* genes are constitutively expressed. *CDT-1* is a novel gene and no genes with similar sequences have been found in other plant species by *in silico* analysis of available databases. *CDT-1* occurs in many copies in the genome of *C. plantagineum* (Furini *et al.*, 1997). *CDT-1* has features of a short interspersed element (SINE) retrotransposon, and footprints of transposition events were found in the genome of *C. plantagineum* (Furini, personal communication). The mechanism, by which *CDT-1* activates genes, is not understood. The mRNA has only

Structure of the *CDT-1* transcript

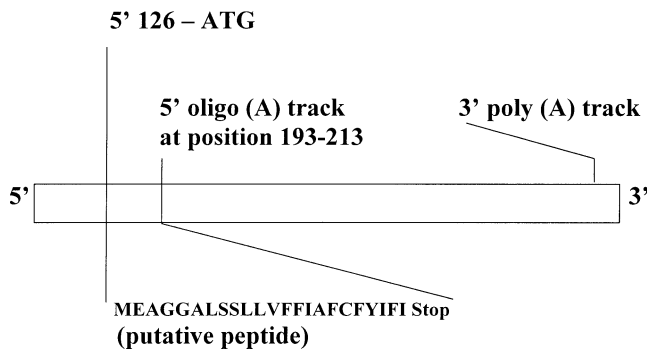


FIG. 2. The figure shows the predicted organization of the *CDT-1* transcript according to Furini *et al.*, 1997. Indicated as amino acid sequence is the putative polypeptide with an ATG 126 nucleotides downstream of the 5' end, the other features are a stretch of adenine residues between nucleotides 193 and 213 and a poly(A) track at the 3' end.

one putative open reading frame which has the capacity to encode a 22 amino acid polypeptide (Fig. 2). The existence of the hypothetical polypeptide could not be demonstrated so far, and all experimental evidence points to the possibility that *CDT-1* functions as a regulatory non-coding RNA (Furini and Bartels, unpublished). Recently other members of the gene family have been isolated and characterized (Smith-Espinoza *et al.*, 2005). It will be intriguing to know whether the *CDT-1* gene is present in species closely related to *C. plantagineum*. A function of *CDT-1* as regulatory RNA is supported by observations made from studying bacteria. It was shown that some short (70 to 120 nucleotides long) bacterial RNA species can function as sensors for environmental conditions (Weisberg and Storz, 2002). The RNAs control gene expression post-transcriptionally and relay information about the outside temperature to the translational machinery. The RNAs form a secondary structure which blocks certain ribosome binding sites at low temperature. If the temperature increases, the secondary structure of the RNAs is resolved, the ribosomes attach to the Shine-Dalgarno sequence and protein synthesis starts. In analogy *CDT-1* RNAs could assume a different structure during water depletion and allow transcription of dehydration-induced mRNAs *e.g.*, by interacting with chromatin components.

Carbohydrate metabolism

Another unique and unusual feature of *C. plantagineum* is the carbohydrate metabolism during the dehydration and rehydration cycle. *C. plantagineum* contains high amounts of the unusual C8 sugar octulose in fully hydrated leaves (*ca.* 400 mg octulose g⁻¹ lyophilized leaf tissue). Upon dehydration the octulose level declines and sucrose accumulates to similar high levels and the process is reversed during rehydration (Bianchi *et al.*, 1991). The accumulation of high levels

of sucrose in vegetative leaves during dehydration seems to be another feature characteristic for resurrection plants studied so far. The biochemical role of sucrose in this process is not understood. A protective role of sucrose is supported by *in vitro* studies which showed that a wide range of biomolecules are less susceptible to denaturation when dehydrated in the presence of sugars (Crowe *et al.*, 1992).

Conclusions and outlook

The distribution of resurrection plants within the plant kingdom suggests independent acquisition of desiccation tolerance in several plant species. Desiccation tolerance became an advantage for plants during the transition from life in water to land plants. All molecular studies performed with resurrection plants so far indicate that resurrection plants have not obtained novel genes, but that nearly all gene products, which are expressed in correlation with desiccation tolerance, have counterparts in non-tolerant plant species. The exception are the *CDT-1* and -2 genes (Furini *et al.*, 1997; Smith-Espinoza *et al.*, 2005) isolated from *C. plantagineum*. Apart from the *CDT-1* gene, comparisons of protein coding sequences of dehydration-induced genes suggest that many genes which are abundantly expressed in vegetative tissues in resurrection plants are part of the gene expression program of a non-tolerant plant, but the abundant expression is restricted to the seed maturation phase when the embryo becomes desiccation tolerant. The expression of the genes declines with the onset of germination and thus the expression is strictly developmentally regulated. Prominent example are the different LEA protein encoding genes (see above). The similarities are often not restricted to the protein coding regions of the genes but extend to the presence of similar regulatory *cis*-promoter elements illustrated *e.g.*, by ABA response elements present in many seed specific genes (Bartels and Sunkar, 2005; Ingram and Bartels, 1996; Leung and Giraudat, 1998; Shinozaki and Yamaguchi-Shinozaki, 2000).

The conclusions drawn here rely on very restricted genomic information. Identification of genes has mostly been based on the identification of highly conserved protein coding orthologs. It is becoming clear from the analysis of desiccation tolerance that desiccation tolerance is a complex trait, in which many genes are involved, relies on the biochemical function of the genes and on how gene transcription is regulated. In the light of the importance of gene regulation and in the context of the discovery of the *CDT-1*-genes attention should be drawn to the investigation of non-protein-coding sequences. So far it has not been considered that, in addition to the conservation of protein coding sequences, desiccation tolerance is crucially linked to the correct functioning and recognition of *cis*-regulatory sequences. These regulatory sequences may even lie at considerable distance from the protein coding sequences whose expression they control. Identifying the linkage between *cis*-regulatory sequences

and protein coding sequences will lead to a better understanding of how co-ordinated expression of dehydration-relevant genes has been achieved and how it can be exploited in manipulating desiccation tolerance. It can be expected that, as more analyses of genome sequences become available, particularly from closely related species with different tolerance levels, our insight into molecular aspects of evolution of desiccation tolerance will be greatly enhanced.

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